



Mu Sigma

DO THE MATH

Vertical-Writeup

Pharmaceutical Industry

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1. Overview of Pharmaceutical Industry

1.1 Introduction of pharmaceutical industry

The pharmaceutical industry is at the heart of healthcare, dedicated to the discovery, development, and commercialization of medicines. These medicines are crafted to cure diseases, prevent illnesses through vaccines, and alleviate symptoms like pain, ultimately enabling people to lead healthier and longer lives.

The healthcare sector can be broadly categorized into four key segments:

Pharmaceuticals: Focused on the creation of medicines and drugs.

Medical Technology: Involves the development of medical tools and equipment, such as X-ray machines, pacemakers, and even health apps.

Biotechnology: Utilizes living cells and biological processes to develop innovative treatments.

Healthcare Services: Comprises the professionals and facilities that deliver care, including doctors, nurses, and hospitals.

1.2 History of Pharma

The pharmaceutical industry has developed over centuries, moving from natural remedies to advanced modern medicines:

Before the 19th Century: Medicines were prepared using natural ingredients like herbs, minerals, and animal products. Treatments were based on cultural practices and trial-and-error, with little scientific validation.

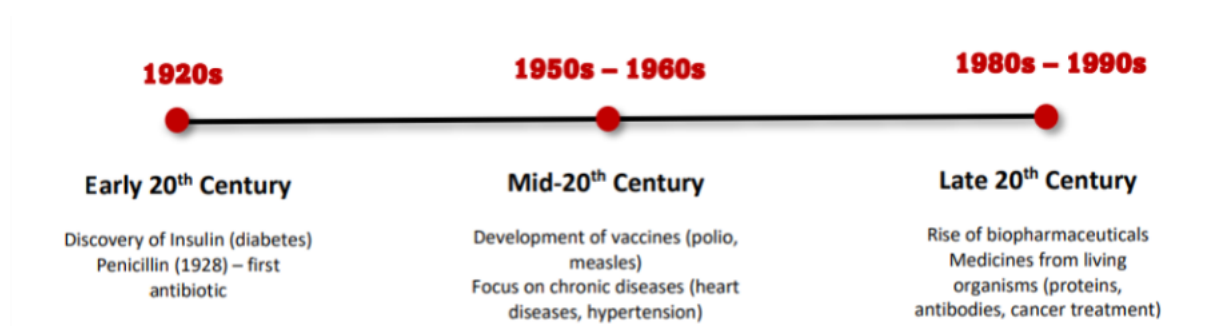
19th Century: The discovery of morphine in 1804 marked a shift toward scientific drug development. Major pharmaceutical companies like Pfizer (1849) and Bayer (1863) were founded, initiating industrial-scale medicine production.

Early 20th Century: The 1906 Pure Food and Drug Act introduced the first formal drug safety regulations in the U.S. Penicillin, discovered in 1928, became the first widely used antibiotic, transforming infection treatment.

Mid-20th Century: The 1950s–60s saw rapid drug innovation, with breakthroughs in treating heart disease, infections, and mental health.

Late 20th Century: Biotechnology emerged, enabling the development of genetically engineered drugs and targeted therapies. Pharma companies began focusing on chronic conditions like diabetes, cancer, and autoimmune diseases.

21st Century: Personalized medicine became a reality, with treatments tailored to individual genetic profiles. Artificial intelligence began playing a major role in drug discovery, and attention shifted toward rare diseases and orphan drugs.



1.3 Why is Pharma important?

Advancements in Healthcare: The pharmaceutical industry has dramatically transformed healthcare by developing groundbreaking therapies and medications that tackle complex diseases, boost survival rates, and improve quality of life. Ongoing innovations, such as personalized medicine, immunotherapies, and advanced biologics, are reshaping treatment approaches across a wide range of medical conditions.

Economic Contributions: Beyond the health benefits, pharmaceuticals significantly impact global trade and economic growth. They contribute to national GDPs, generate high-skilled employment opportunities, and propel research and technological advancements. Pharmaceutical companies are often at the forefront of R&D investment, driving innovation and nurturing a vibrant ecosystem of scientific discovery and application.

2. Drug Categories

Drugs are now categorized according to their commercial value, regulatory status, and market dynamics in addition to their intended use and mechanism of action.

2.1 Blockbuster Drugs

Medicine is considered a blockbuster if its yearly global sales exceed \$1 billion. These are crucial sources of income for big pharmaceutical corporations and are typically used to treat common or chronic illnesses.

Key Traits

- Provide a wide range of indications, such as cardiovascular disorders and cancer.
- Usually shielded by exclusivity or patents.

Example

- Opdivo – A PD-1-targeting cancer immunotherapy.

2.2 Speciality Drugs

Specialty medications are made to treat uncommon, chronic, or complicated illnesses. They are typically given under a doctor's supervision, are frequently biologics, and may need specific storage.

Key Trait

- Aim for severe illnesses including cancer, autoimmune diseases, or genetic abnormalities.

Example

- **Breyanzi** and **Abecma** – CAR-T cell therapies for blood cancers.

2.3 Generic Drugs

Generic medications are less expensive than name brand ones whose patents have expired.

Key points

- The same active components and dosage as the original medication.
- Reduce healthcare expenses and improve access.

For instance, generic erosion is already affecting its medication Revlimid, which affects portfolio planning and revenue projections.

2.4 Over the Counter (OTC) Drugs

Common, self-managed ailments like colds, headaches, and allergies are treated with over the counter (OTC) medications, which are accessible without a prescription.

Key traits

- Safe for general public use when used as directed by the label.
- High volume at low pricing.
- Widely accessible at pharmacies that sell retail drugs.

2.5 Orphan Drugs

Orphan medications are created for uncommon illnesses that frequently have few available treatments.

This refers to illnesses that impact less than 200,000 persons in the United States.

Key traits

- Target important yet specialized markets.
- Take advantage of incentives like as tax breaks and prolonged market exclusivity.

BMS has specifically invested on orphan medications, particularly in the fields of oncology and hematology. For instance, reblozyl is authorized for the treatment of myelodysplastic syndromes (MDS) and beta thalassemia, two uncommon types of anemia.

3. Current Industry Landscape

3.1 Innovation is Moving Fast — and in New Directions

The speed at which science is developing is unprecedented. R&D efforts are still led by therapeutic fields such as immunology, neurology, oncology, and rare illnesses, but the emphasis is now on customized and precision medicine. Technologies like protein degraders, bispecific antibodies, cell and gene therapies, and mRNA-based medicines are revolutionizing the way diseases are treated and, in certain situations, cured.

3.2 Pricing, Regulation, and Access Are Under the Microscope

Governments and payers around the world are advocating for greater authority over drug pricing and reimbursement schemes. Pharmaceutical businesses are facing increasing pressure to improve affordability and justify value.

To improve fair access, especially in underdeveloped markets, BMS has reacted proactively by creating global access programs, putting value-based contracts into place, and collaborating with healthcare systems.

3.3 M&A and Pipeline Prioritization Are Driving Portfolio Strategy

Leading pharmaceutical companies are actively restructuring their portfolios to preserve innovation and competitiveness. This has resulted in a series of strategic acquisitions for BMS, such as Rayze Bio (radiopharmaceuticals), Mirati Therapeutics (oncology), and Karuna Therapeutics (neuroscience).

3.4 Digital Tools and AI Are Transforming Drug Development

The development and discovery of new medications is increasingly influenced by technology. Real-world evidence (RWE), machine learning, and artificial intelligence (AI) are accelerating the design of clinical trials, enhancing target selection, and increasing the effectiveness of R&D. To improve decision-making and shorten development schedules, BMS has used these tools, which is a crucial edge in the cutthroat industry of today.

4. Key Trends and Developments in Pharmaceutical Industry

Key trends and developments in the pharmaceutical industry are crucial to understand as they shape the future of healthcare and medicine.

4.1 Blockchain Technology

Blockchain is a decentralized digital ledger that records transactions across several computers in a way that the recorded transactions cannot be altered. Companies use unique identifiers, verify the authenticity of each drug, prevent the circulation of counterfeit medicines. Example: Used for supply chain transparency and ensuring the authenticity of pharmaceutical products, preventing fraudulent medicines.

4.2 Artificial Intelligence (AI) and Machine Learning (ML)

AI and machine learning involve the development of computer systems that can perform tasks typically requiring human intelligence, such as decision-making and problem-solving. Example: AI is used in drug discovery to analyse large datasets more efficiently and predict potential drug candidates. ML aids therapeutic target identification by analysing a variety of data sources, including clinical, proteomic and genomic data.

4.3 Personalized Medicine

It is also referred to as precision medicine. Personalized medicine tailors' medical treatment to the individual characteristics of each patient, often using genetic or molecular profiling. It also involves personalizing early detection strategies, intensive personalized health monitoring etc. to determine health status change. Example: Genomic-based treatments that target specific genetic mutations in diseases such as cancer.

4.4 Telemedicine and Digital Health technologies

Telemedicine refers to the use of digital information and communication technologies to access health care services remotely. Advances in DHTs, including electronic sensors, computing platforms, and information technology, provide new opportunities to obtain clinical trial data directly from patients. Example: Virtual consultations and remote monitoring of patients to manage chronic conditions and ensure medication adherence.

4.5 Biopharmaceutical Innovations

Involves the creation of medical drugs produced using biotechnology, often from living organisms. Example: Development of monoclonal antibodies and gene therapies for targeted treatment of diseases like cancer and genetic disorders.

4.6 Decentralized Clinical-Trials

Decentralized clinical trials refer to clinical trials where some or all the activities occur at a location other than traditional clinical trial site. Example: Can be performed at larger scale, spanning across centres and countries to enhance diversity and robustness of trials. DCTs use digital tools and platforms to collect data remotely, allowing for immediate access and continuous monitoring.

5. Drug Development Lifecycle

5.1 Discovery and Development

a. Discovery

The discovery stage is the first step in bringing a new drug to market. Typically, researchers evaluate compounds to determine which could be candidates for development as potential medical treatments. Several methods are used in the discovery process. These include:

- Testing many molecular compounds to find if they offer possible benefits against diseases.
- Using new information about diseases to design products that could stop or reverse the effects of the disease

b. Development

Involves conducting experiments to gather information on:

- Pharmacokinetics of the drug as in how it is absorbed, distributed, metabolized and excreted. Its potential benefits, best dosage, side effects or adverse effects.
- How it interacts with other drugs and treatments and its effectiveness as compared with similar drugs.

5.2 Preclinical Research

Preclinical research or non-clinical studies are conducted to assess the toxicity and activity of new compound in animals (in vivo testing) and human cells (housed in laboratory vessels, in vitro testing). Toxicity studies involve testing new compounds in animals at increasing doses to see how they respond. If the new compound is found to be safe and effective in cells, all findings of the preclinical research are submitted to regulatory authorities for permission to conduct clinical testing in humans.

5.3 Manufacturing of the drug according to GMP standards

The drug developer is required to provide data on the production method and testing results as the compound is developed. Initially produced in small bench side batches, the drug undergoes a scale-up process using Good Manufacturing Practices (GMP). Future manufacturing suitability, commercial viability, and cost-effectiveness must be considered.

5.4 Clinical Research

Clinical research involves studies or trials conducted with human participants. As the developers create the clinical study, they will think about their goals for each of the various Clinical Research Phases and initiate the Investigational New Drug Process (IND), the application for clinical testing in humans, which is submitted to the FDA.

a. Designing Clinical Trials

Researchers develop clinical trials to address research queries concerning a medical product. These trials adhere to a detailed study plan, known as a protocol, which is created by the researcher or the manufacturer.

b. Phases of Clinical Trials

- ▶ Phase 1: 20- 100 healthy volunteers or people with disease/ condition to determine safety and dosage.
- ▶ Phase 2: Up to several hundred people with disease/ condition to determine efficacy and side effects.
- ▶ Phase 3: 300 to 3000 volunteers who have disease or condition to determine efficacy and monitoring of adverse reactions.
- ▶ Phase 4: Studies are conducted after a compound has been approved, for the primary purpose of post marketing surveillance.

5.5 Drug application approval process

Regulatory authorities evaluate all the available data for safety and efficacy prior to approval. In the U.S., submission of New Drug Application (NDA) or Biologics License Application (BLA) is required.

If the FDA approves the new compound, the drug developer can market it and scale up manufacturing from clinical trial batches to commercial size.

6. Post Market Surveillance (Pharmacovigilance)

6.1 What is Post Market Surveillance

Post Market Surveillance refers to the systematic process of monitoring pharmaceutical products after they receive regulatory approval and enter the market. Unlike clinical trials, which happens in controlled settings with limited groups, PMS collects real-world data from various patients, treatment environments, and usage trends. It plays a vital role in identifying rare, long-term, or population-specific adverse effects that may not be evident during pre-market testing. PMS is a cornerstone in pharmacovigilance, ensuring that drug safety and efficacy are continuously evaluated throughout the product lifecycle.

6.2 Why PMS Matters in 2025

Today's pharmaceutical landscape, PMS is more critical than ever. The rise of accelerated approvals, personalized medicines, and complex biologics has heightened the need for strong post-approval monitoring. Drugs are now reaching patients faster, often with little long-term data. This makes real-world surveillance vital in maintaining public trust. It also supports adaptive regulatory frameworks that rely on ongoing data to make informed decisions about labelling, usage restrictions and market continuation.

6.3 Monitoring Adverse Drug Reactions (ADRs)

Adverse Drug Reactions are a central focus of PMS, and in 2025, their detection has become more sophisticated, and data driven. Pharmaceutical companies and regulators use a mix of traditional and digital tools to track and evaluate ADRs:

- ▶ **Spontaneous reporting systems** like VigiBase and EudraVigilance are still essential, but they now use AI for signal detection.
- ▶ **Real-world data** from electronic health records (EHRs), insurance claims, and wearable devices allow for ongoing passive monitoring.
- ▶ **Patient-reported outcomes** gathered through mobile apps and online portals boost reporting rates and highlight subtle side effects.
- ▶ **Natural Language Processing (NLP)** tools shift through social media, forums, and medical literature to spot risks.

6.4 Lifecycle Risk Management Plans (RMPs)

Risk Management Plans in 2025 are dynamic and updated regularly. They describe strategies to find and reduce risks, regularly. They describe strategies to find and reduce risks, especially for high-risk therapies. Regulators now require regular revisions based on real-world evidence and patient feedback. This ensures that safety measures stay relevant and responsive.

6.5 Labelling Updates, Warnings & Product Recalls

Labelling revisions and product recalls are key outcomes of effective PMS. When new safety signals emerge, regulators and manufacturers must act swiftly to update product information and protect patients. In 2025, the process is faster and more transparent thanks to digital traceability tools:

- ▶ **Labelling Updates** may include new contraindications, dosage changes, or black box warnings.
- ▶ **Blockchain systems** ensure secure tracking of drug batches and allow for quick recall actions.
- ▶ **Digital platforms** notify prescribers, pharmacists, and patients in real time.
- ▶ **Global standardization** of recall procedures ensures consistency across markets and reduces disruption.

6.6 Role of AI & Real-World Evidence (RWE)

AI and real-world evidence have made PMS more predictive and responsive. Algorithms detect safety signals early, while real-world data from EHRs, claims, and wearables support faster risk assessments. Together, they help regulators and pharma companies act swiftly and personalize safety strategies.

7. Pharmaceutical Value Chain

The pharmaceutical industry operates through a network of stakeholders where **products, payments, rebates, and promotions** flow simultaneously. Understanding these interactions ensures **operational efficiency, financial transparency, and regulatory compliance**.

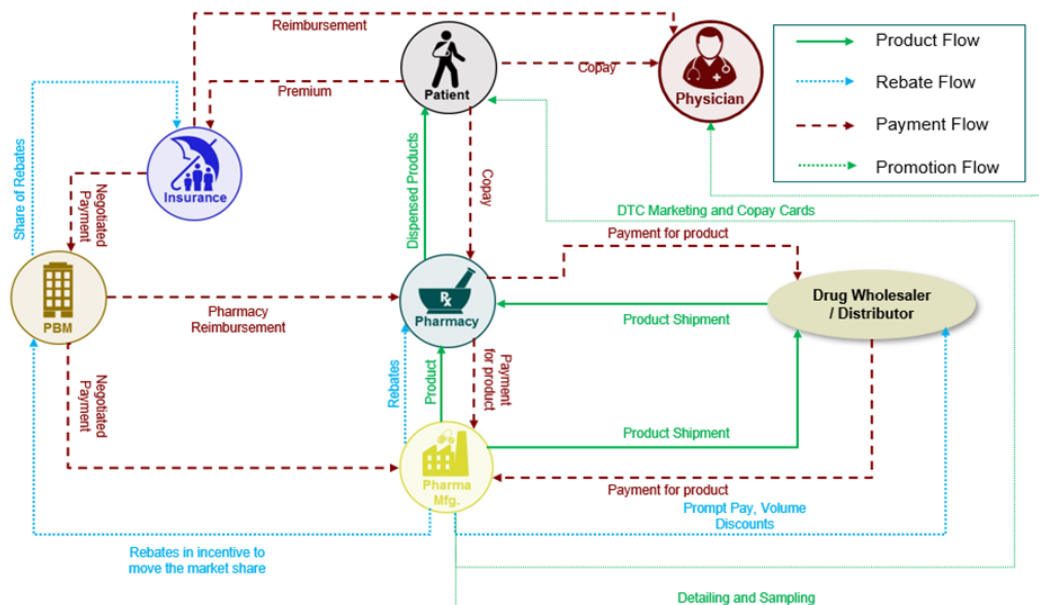


Figure 1. Stakeholder Interaction

7.1 Key Stakeholders

Stakeholder	Role
Patient	Pays premiums and copays; consumes medicines.
Physician	Prescribes drugs based on medical need and awareness.
Pharmacy	Dispenses medicines and manages reimbursements.
Wholesaler	Distributes drugs efficiently to pharmacies.
Insurance	Covers costs and manages reimbursements.
PBM	Negotiates drug prices and rebates.
Manufacturer	Produces medicines and runs marketing and rebate programs.

7.2 Product Flow

Medicines move through a structured supply chain:

Pharma Manufacturer → Wholesaler/Distributor → Pharmacy → Patient

- Manufacturers produce drugs and supply them to wholesalers.

- Wholesalers distribute them to pharmacies.
- Pharmacies dispense the prescribed medicines to patients.
- In some cases, manufacturers directly ship products to large pharmacy chains or hospitals.
This ensures quality, accessibility, and traceability.

7.3 Payment Flow

Money moves in multiple directions to balance affordability and profitability:

- **Patient → Insurance:** Premium payments.
- **Insurance → Patient:** Reimbursement for eligible costs.
- **Patient → Pharmacy:** Copay for medicines.
- **Insurance → Pharmacy:** Reimburses insured drug costs.
- **Pharmacy → Wholesaler → Manufacturer:** Payments for purchased inventory.
- **PBM ↔ Insurance & Manufacturer:** Negotiates prices, rebates, and reimbursement structures.

7.4 Rebate Flow

Rebates maintain competitive pricing and incentivize stakeholders:

- **Manufacturer → PBM:** Rebates for promoting specific drugs.
- **PBM → Insurance:** Shares rebate savings.
- **Manufacturer → Wholesaler:** Volume or early payment discounts.
- **Pharmacy → Wholesaler:** May receive prompt pay discounts.

7.5 Promotion Flow

Promotions drive awareness among doctors and patients:

- **Manufacturer → Physician:** Detailing, sampling, and co-pay assistance.
- **Manufacturer → Patient:** Direct-to-consumer marketing and discount cards. These activities encourage prescriptions and patient engagement while remaining under strict compliance.

8. Challenges and Future Prospects

8.1 The Challenges

The Innovation Bottleneck

The journey from molecule to medicine is a marathon. For every 10,000 compounds screened, only one might reach the pharmacy shelf. The attrition rate is brutal many candidates fail due to toxicity, poor efficacy, or unexpected side effects. Even with cutting-edge tools, predicting how a drug behaves in the human body remains a complex puzzle. The cost of failure is high, both financially and emotionally, especially for diseases with no current cure.

Regulatory Complexity

Each country has its own regulatory fortress. The FDA in the U.S have distinct requirements. Companies must navigate thousands of pages of documentation, conduct region-specific trials, and respond to endless queries. While these systems protect public health, they also slow innovation. Harmonizing global regulations remains a distant dream.

Pricing Pressures

Pharma companies walk through a tight rope between innovation and affordability. Developing a new drug costs billions, but patients and governments demand lower prices. In some regions, price caps and reimbursement restrictions threaten the viability of launching new therapies.

Global Supply Chain Fragility

The COVID-19 pandemic was a wake-up call. Lockdowns, export bans, and raw material shortages disrupted the flow of essential medicines. Even today, geopolitical tensions and climate events continue to destabilize supply chains. Pharma companies are now rethinking their sourcing strategies, investing in local manufacturing, and building redundancy into their logistics.

Talent & Technology Gaps

As the industry embraces AI, data science, and genomics, the demand for hybrid talent scientists who understand both biology and algorithms is soaring. Traditional pharma firms struggle to attract this new breed of professionals, who often prefer startups or tech giants. Upskilling the existing workforce is urgent but challenging.

8.2 The Future Prospects

AI-Powered Discovery

Artificial Intelligence is no longer a buzzword, it's a game-changer. AI models can simulate molecular interactions, predict toxicity, and even design new compounds. Companies like Insilico Medicine and DeepMind are proving that algorithms can accelerate discovery and reduce costs. In the future, AI might even personalize drug regimens based on a patient's genome and lifestyle.

Precision Medicine

The era of "one-size-fits-all" is fading. Precision medicine uses genetic, environmental, and lifestyle data to tailor treatments. In oncology, for example, therapies are now designed to target specific mutations. This approach not only improves outcomes but also reduces side effects. As sequencing becomes cheaper, precision medicine will expand to cardiology, neurology, and beyond.

Digital Health Integration

Smartwatches, mobile apps, and remote sensors are transforming how drugs are tested and monitored. Clinical trials can now be decentralized, with patients participating from home. Real-world data collected from devices helps refine treatments and detect side effects early. Pharma companies are partnering with tech firms to build digital ecosystems around their products.

Sustainable Pharma

Environmental sustainability is becoming a core priority. Green chemistry aims to reduce hazardous waste in drug synthesis. Companies are investing in biodegradable packaging, energy-efficient plants, and carbon-neutral logistics. Regulators are also pushing for eco-friendly practices, making sustainability a competitive advantage.

Global Collaboration

The success of mRNA vaccines showed what global collaboration can achieve. Pharma companies, governments, and academia are now working together more than ever. Shared databases, open-source research, and cross-border trials are accelerating innovation. In the future, we may see global task forces tackling pandemics, rare diseases, and antimicrobial resistance.

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